Predictive and Reactive Robot Navigation in Shared Urban Pedestrian Spaces

Evidence from a MetaUrban Pilot

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UCMP-MB · Week 3 · Option 1: Mini Research Proposal

1. Introduction and Research Question

1.1 Urban Context and Theoretical Motivation

A service robot travelling to an inspection point must share crossings and pavements with people. Its arrival time alone cannot describe whether that journey fits the public space. Street furniture can constrain movement, while pedestrian motion changes when two users compete for the same location.

Urban informatics connects spatial data and movement records with analytical methods to examine shared public space. A network perspective on cities emphasises movement and interaction [1]. At pavement scale, this motivates examining street form together with time-varying encounters. Here, spatial structure constrains possible actions; recorded trajectories reveal interactions; comparisons inform robot movement decisions. Accessibility motivates evaluating task completion alongside the spatial and temporal costs of sharing pavements.

Social-navigation research also distinguishes physical safety, comfort and understandable behaviour [3]. A robot may avoid recorded contact yet remain close to someone or obstruct their passage. Consequently, this study assesses arrival, contact, waiting and proximity separately to describe different aspects of sharing space.

1.2 Research Question and Objective

How does short-horizon motion prediction affect contact-free arrival and robot waiting in shared urban pedestrian spaces, and which encounter or layout conditions limit its usefulness?

Reactive control responds to current conditions; prediction estimates how encounters may change over the next few seconds. We test that added information with a shared route in one intersection family, without claiming general effects of street design.

1.3 Pilot Work and Proposed Follow-up

The completed pilot implemented a route-conditioned forecast and speed-selection module, adapted comparison methods, and evaluated 160 formal runs after 68 development runs. The matched comparisons characterise completion, contact and waiting across pedestrian encounters and identify conditions for further testing. The proposed follow-up is to test a bounded-detour trigger when waiting cannot resolve static blockage. That extension is not yet implemented.



Figure 1. MetaUrban views at departure and after arrival and physical stopping in a development case. Crossings, pavement and street objects locate the inspection task. RGB images document execution; controllers use numerical states.

2. Urban Simulation Data and Spatial Units

2.1 Data Source and Urban Scenarios

MetaUrban provides three-dimensional urban micromobility settings [2]. We use its public small asset set, an intersection, street furniture and a wheelchair-shaped wheeled model with native wheel physics. Controllers use simulator states and a supplied reference route; evaluation focuses on wheeled navigation.

The main test combines ten newly evaluated scenes with two pedestrian counts, five and fifteen, giving twenty matched conditions for each of six methods. A separate diagnostic suite tests crossing, receding, approaching and stationary pedestrians, with two new conditions per encounter and five methods. These forty single-person runs are not pooled with the 120 main runs. Sixty-eight development runs preceded both tests.

Native crowd paths use Optimal Reciprocal Collision Avoidance (ORCA), an approach based on shared avoidance responsibility [7]. Their realised motion here does not establish responsive human behaviour. Diagnostic pedestrians instead follow prescribed trajectories. Controllers observe their current positions and derived velocities, not the future scripts.

2.2 Variables, Spatial Units and Data Boundaries

The data connect urban conditions to measurable decisions as follows. A footprint is the ground-plane area of an actor’s collision box. Crowd size is an agent count, not a density in people per square metre. Pavement and crossing context is visible in the scene, but their widths were not systematically varied or evaluated as causal factors.

Table 1. Urban context, recorded variables and analytical roles.

Urban attribute	Recorded variable and unit	Role in the study
Route and task location	Reference polyline, robot position and endpoint distance (m)	Identify progress and actual waypoint arrival
Street furniture	Static object footprints and predicted gap (m)	Inspect persistent stops near the reference route
Pedestrian movement	Position (m), derived velocity (m/s), count and encounter type	Describe where and when trajectories conflict
Sharing a location	Contacts, stop-command time (s), proximity exposure (%)	Compare progress with observable interaction costs

Coordinates are local metres without a geographic reference. One 0.1-second decision is the observation unit; one episode is the outcome unit. Scene identity and crowd count define a matched comparison. Both crowd conditions share the same scene identity, so twenty conditions do not represent twenty independent urban layouts.

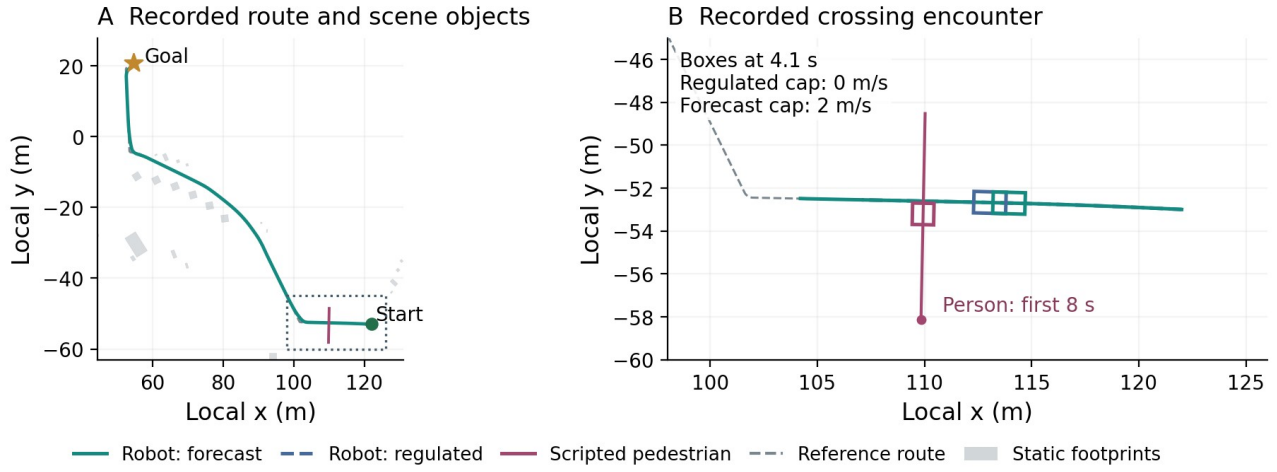


Figure 2. Recorded development crossing encounter. A shows the reference route, executed forecast trajectory, target and static footprints. B shows observed positions and commands, the pedestrian’s first eight seconds and body outlines at 4.1 seconds. The figure uses local simulator coordinates and recorded object geometry.

3. Methodology and Evaluation Design

3.1 Pilot Module and Comparison Methods

The pilot modification is the measured-velocity forecast module. It selects speed while a common tracker follows the route. We also implemented the controls and local baseline adaptations in Table 2. All receive the same full route, robot state, pedestrian positions and derived velocities, and static footprints, with shared motion limits, braking and static protection.

Pure Pursuit (PP) chooses steering towards a point ahead on the route [4]. Regulated Pure Pursuit (RPP) adds speed regulation around curves and obstacles [5]. The Dynamic Window Approach (DWA) searches locally achievable motion commands [6]. The RPP-inspired and DWA-style versions below are local adaptations, not full reproductions of the published controllers.

Table 2. Roles in our completed pilot. The final column explains what each comparison tests.

Method	Study role	Decision and comparison purpose
Forecast: measured velocity	Pilot modification	Predict bodies along candidate route motions; extrapolate measured pedestrian velocity and choose speed or stopping.
PP + static guard	Baseline	Track the route with basic static protection; test a simple backbone [4].
RPP-inspired	Baseline	Regulate speed using curves and nearby objects; compare a regulated tracker [5].
DWA-style	Baseline	Search local speed and steering choices; test lateral flexibility [6].
Distance yield	Simple control	Slow using current pedestrian gap; test whether prediction is needed.
Forecast: zero velocity	Ablation control	Same module with predicted pedestrian velocity set to zero; isolate that input. Actual people still move.

Our module compares normal progress, slower motion and stopping over 1.5 seconds, following the reference route in each prediction. Oriented body footprints test whether a robot and a person may share space at similar times. This speed-selection layer is the implemented modification; it does not introduce a new global route.

The zero-velocity ablation changes only pedestrian-velocity input. It tests that input’s value, whereas comparison with RPP also includes other controller choices. Because a prescribed person may approach a stationary robot, stopping is evaluated alongside forward-motion candidates. ORCA generates crowd motion; it is not an additional robot baseline here.

3.2 Outcomes and Fair Comparisons

Strict arrival requires both the simulator’s arrival flag and an endpoint distance of at most 2 m. Contact-free arrival additionally excludes simulator-reported contacts with pedestrians, objects, buildings and sidewalk boundaries. Pedestrian-contact counts are reported separately because an episode can reach the goal after contact. Stop-command duration measures robot waiting decisions, not pedestrian delay.

Proximity exposure is the observed time fraction with a planar robot–pedestrian footprint gap below 1 m, sampled every 0.1 seconds. This geometric proxy is not measured comfort. Its denominator ends when the episode ends, so exposure must be interpreted alongside completion and contact outcomes.

Each method starts from the same physical scene and has the same sixty-second budget. Routes and start-to-goal distances must both be at least 5 m; invalid tasks would be retained without replacement. All twenty main conditions passed this check. Parameters were fixed after development and before either test, with failed and contact episodes retained.

Analysis uses descriptive paired counts and compares travel time only on conditions completed by both methods. Shared scene identities are respected when interpreting repeated crowd conditions. No significance or equivalence claim is made. Detailed software versions, scene identifiers, engineering thresholds and supplementary uncertainty calculations are documented separately.

4. Preliminary Results and Interpretation

4.1 Preliminary Comparison and the Velocity Ablation

Table 3 summarises the main test. The measured-velocity forecast and its zero-velocity ablation each recorded sixteen strict arrivals, thirteen contact-free arrivals and three pedestrian-contact episodes. This pilot therefore found no aggregate completion or contact-free benefit from the velocity input itself.

Table 3. Completed main test: twenty matched conditions per method. Our module is the measured-velocity row; comparison roles are in Table 2. Counts are episodes; time and exposure are episode means.

Method	Strict	Contact-free	Ped. contact	Wait (s)	Near (%)
PP + static guard	16	13	3	9.47	0.46
RPP-inspired	16	13	3	10.38	1.41
Distance yield	16	13	3	9.90	1.27
Forecast: zero velocity	16	13	3	10.31	1.43
Forecast: measured velocity	16	13	3	10.52	1.40
DWA-style	14	12	2	14.55	0.84

Against RPP-inspired control, forecasting was 3.16 seconds faster across the same 16 successful conditions. However, it added 0.28 seconds relative to the zero-velocity ablation on those same conditions. The difference from RPP is specific to that comparison; it establishes neither a general time advantage nor a benefit from pedestrian velocity input.

PP with the common static guard matched the same arrival counts and had lower mean waiting and proximity exposure. Thus, increased predictive complexity was not justified as a default by this sample. None of these comparisons measures people’s perceived comfort or rare-event safety.

4.2 Failure Patterns and Lateral Flexibility

All five route-tracking variants timed out in the same four scene–crowd conditions. Their moving predictions failed the static-footprint buffer while stopping passed. DWA-style completed all four, three without contact, but failed six other conditions completed by forecasting. This indicates a trade-off in lateral control, not universal superiority.

In paired strict-arrival terms, forecast gained 6 conditions and lost 4 relative to DWA-style, a net gain of 2. The contact-free total changed separately from twelve to thirteen, a gain of 1. The two comparisons show why reaching the goal and doing so without contact must be evaluated separately.

Table 4. Separate scripted diagnostics: contact-free arrivals out of two conditions per encounter. One pedestrian is present in each episode.

Encounter	RPP-inspired	Distance	Zero velocity	Measured velocity	DWA-style
crossing	2	2	2	2	1
receding	2	2	2	2	2
approaching	0	0	0	0	1
stationary	0	0	0	0	0

In crossing diagnostics, forecasting averaged no waiting against one second for regulated tracking, but proximity exposure rose from zero to 2.42%. All methods completed both receding cases without contact. The four speed-only methods contacted both approaching pedestrians; DWA-style achieved one contact-free arrival. No method finished either stationary-person task. Waiting alone therefore did not resolve every encounter.

5. Discussion and Proposed Next Study

5.1 What the Preliminary Results Support

The pilot supports a focused follow-up on spatial alternatives. Velocity input left aggregate strict and contact-free arrival unchanged, while persistent stops indicate limits of speed adjustment along a fixed route. Timing differences depend on the comparison method. Static-buffer rejections identify controller-level constraints; assessing physical passability also requires validating geometry and feasible motion.

The urban implication is a conditional design question: when should a public-service robot keep waiting, and when should it consider another nearby path? A faster robot journey is insufficient justification if the alternative transfers obstruction or risk to pedestrians. Those human consequences require additional evaluation.

5.2 Proposed Extension and Follow-up Experiment

Our proposed bounded-detour trigger has not been implemented. It would extend the existing module after a persistent stop caused by static checks, considering left and right waypoint offsets inside a limited local area. Candidates must remain in validated walkable space and preserve footprint buffers; otherwise the robot would keep waiting.

Our follow-up would compare three groups: RPP-inspired tracking (baseline), our existing forecast module (reference), and our module plus the new trigger (proposed variant). Only the third adds the detour. Ten new scene identities, two validated layout families and two crowd sizes give 120 new runs across those groups. The hypothesis is more strict arrivals without fewer contact-free arrivals.

We would first develop the trigger on separate scenes, fix its settings and protocol, then run the matched test and review failures by layout and crowd count. Existing local simulation makes this feasible without training. These 120 runs and any detour benefit are future work; they are not part of the completed 160-run evidence.

5.3 Uncertainty, Ethics and Transferability

The pilot covers one intersection family, a small asset set and ideal simulator states. Simplified wheel dynamics, sampled planar footprints and prescribed pedestrian motion limit transfer. Scene variation does not represent neighbourhood demographics or diverse mobility needs. No causal estimate of pavement width or street design is available.

No real personal observations were collected. Future street-image recording would require appropriate consent or observation procedures, limited retention and accessibility review with affected users. Comfort, predictability and pedestrian inconvenience should be assessed directly [3]. The practical contribution is a reproducible framework for examining urban interaction trade-offs and identifying requirements for further evaluation.

References

[1] Batty, M. (2013). The New Science of Cities. MIT Press.

[2] Wu, W., et al. (2025). MetaUrban: An Embodied AI Simulation Platform for Urban Micromobility. International Conference on Learning Representations (ICLR).

[3] Francis, A., et al. (2025). Principles and Guidelines for Evaluating Social Robot Navigation Algorithms. ACM Transactions on Human-Robot Interaction 14(2), Article 34, 1–65. https://doi.org/10.1145/3700599

[4] Coulter, R. C. (1992). Implementation of the Pure Pursuit Path Tracking Algorithm. CMU-RI-TR-92-01, Carnegie Mellon University.

[5] Macenski, S., Singh, S., Martín, F. & Ginés, J. (2023). Regulated Pure Pursuit for Robot Path Tracking. Autonomous Robots 47, 685–694. https://doi.org/10.1007/s10514-023-10097-6

[6] Fox, D., Burgard, W. & Thrun, S. (1997). The Dynamic Window Approach to Collision Avoidance. IEEE Robotics & Automation Magazine 4(1), 23–33.

[7] van den Berg, J., Guy, S. J., Lin, M. & Manocha, D. (2011). Reciprocal n-body collision avoidance. Robotics Research, 3–19. https://doi.org/10.1007/978-3-642-19457-3_1

Appendix. Generative AI Use

I used ChatGPT and Codex (OpenAI) to assist with literature review, controller and evaluation code, local simulation experiments, metric calculations, plots, and English writing and formatting. I requested checks of arrival criteria and failure cases, questioned the choice of comparison methods, and asked for clearer explanations of the implemented module, results and proposed extension. I reviewed the proposal and requested revisions. I remain responsible for understanding and approving the submission.

User Prompt

Clarify what we actually did. Which modules did we implement and evaluate? What do route following and pedestrian-risk prediction do? Which elements come from existing methods, and which changes were developed for this project? Clearly distinguish completed work from the proposed but unimplemented local-detour extension. **Use appropriate comparisons.** I questioned whether PPO was a suitable main baseline. Explain why we introduced Pure Pursuit, RPP-inspired and DWA-style controllers, together with distance-based yielding and the zero-velocity ablation. Explain what each comparison tests. **Review the code and evaluation logic.** I requested closer inspection of the arrival criteria and failure cases, because a coarse success definition could misrepresent performance. Explain the corrections and checks already completed, using genuine three-dimensional simulator execution and recorded data. **Interpret the results honestly.** The predictive method did not demonstrate an overall advantage. Explain the arrival outcomes and timing comparisons without attributing every difference to pedestrian-velocity prediction. Treat this as a small exploratory assignment rather than a SOTA claim. I**mprove the proposal's argument and readability.** Connect the research question, urban spatial conditions, data, variables, spatial units, methods and practical implications. Use clear sections, explanatory transitions and full names for technical abbreviations. Move detailed software versions and engineering parameters outside the main text.**Keep the presentation appropriate for the assignment.** Retain a readable five-page main proposal, including figures and references. Use black headings and balanced spacing. Preserve previous proposal versions and experimental records.**Keep the AI-use record concise and accurate.** Focus on assistance with code changes, experimental comparisons, result checking and document revision.

AI Response

The completed work is a small, training-free MetaUrban pilot. It provides comparative evidence and failure analysis, but does not establish an overall advantage for predictive navigation.

1. **Implemented work.** Route following uses the reference route to select a point ahead, steer towards it and regulate speed around turns and the destination. Earlier corrections also addressed braking and stopping. The pedestrian-risk module evaluates candidate speeds, including stopping, over a 1.5-second horizon. It predicts motion along the route and compares robot and pedestrian body footprints. Pure Pursuit, regulated tracking, motion prediction and local command search are established ideas. This project implemented their local integration, risk-based speed selection, comparison variants and evaluation pipeline; it does not claim to have invented those ideas. The robot uses native three-dimensional simulator physics and privileged numerical states, not camera-based perception or humanoid locomotion.

2. **Comparison methods.** A pretrained Proximal Policy Optimization (PPO) policy alone would mix policy-transfer limitations with changes to route information and control. The final classical suite therefore uses shared route/state inputs and does not execute that neural policy. Pure Pursuit (PP) with a static guard tests a simple tracker. Regulated Pure Pursuit (RPP)-inspired control tests reactive speed regulation. The Dynamic Window Approach (DWA)-style controller tests local speed and steering flexibility. These are local adaptations, not official reproductions. Distance-based yielding asks whether current distance is sufficient. The zero-velocity ablation keeps the forecast procedure but sets predicted pedestrian velocity to zero, isolating that input while actual pedestrians still move.

3. **Evaluation corrections and checks.** Strict arrival now requires both the native arrival flag and an endpoint distance of at most 2 m. Pre-step checks require route length and start-to-goal distance of at least 5 m. Contact-free arrival additionally excludes recorded pedestrian, object, building and sidewalk contacts. Failure and contact episodes remain in the analysis. Pedestrian contacts include both relevant native flags because pedestrian bodies can carry a vehicle tag. Existing acceptance checks compared initial scenes, replayed controller decisions and checked recorded geometry. A subsequent audit independently recomputed outcomes and metrics from 73,885 recorded decisions across 160 formal runs. Genuine simulator videos were checked against physics-only trajectories. These were automated checks with AI assistance, not new experiments in this revision.

4. **What the results show.** After 68 development runs, the completed tests comprised 120 main runs and 40 separately reported single-pedestrian diagnostic runs. In the main test, PP, RPP-inspired, distance-based yielding and both forecast versions each achieved 16/20 strict arrivals and 13/20 contact-free arrivals. DWA-style achieved 14/20 and 12/20. Forecasting was 3.16 seconds faster than RPP on their 16 jointly successful conditions, but 0.28 seconds slower than its zero-velocity ablation on those same conditions. The RPP comparison therefore does not demonstrate a benefit from measured pedestrian velocity or general efficiency superiority. Waiting time measures stop commands; proximity exposure is a geometric proxy, not human comfort.

5. **Proposed work and uncertainty.** The bounded-detour trigger remains unimplemented. It would consider limited left/right waypoint offsets after persistent stopping caused by static checks, preserve footprint buffers and remain within validated walkable space. Otherwise, the robot would keep waiting. The proposed 120-run comparison would test RPP-inspired control, the existing forecast module and that module plus the trigger. This is our follow-up experiment, not an existing result. Approximate geometry, limited layouts and non-responsive or scripted pedestrians constrain interpretation; a controller's rejection of motion does not prove a passage physically impassable.

6. **Proposal revisions.** The five-page main document connects shared urban space, street furniture and pedestrian encounters to recorded variables, local spatial units, matched comparisons and public-space implications. It defines abbreviations, separates methods from findings, explains negative results and distinguishes completed work from hypotheses. Detailed software and engineering settings remain in optional reproduction notes. Headings are black, and figures, tables and seven references remain within the five-page main limit.

7. **AI record and preservation.** The expanded appendix includes your complete question, this complete response and three original screenshots concerning implementation, comparison choices and review requests. Its short disclosure focuses on code assistance, experimental checks and document revision; it omits the earlier topic-selection discussion. Historical screenshots are identified as such. Previous proposals, frozen code and experimental records are retained. No new training or simulation was needed.

Updated files: [PDF](/data/private/user7/projects/Tempfile/UCMP-MB/week3_assignment/report/Week3_Streetscape_Research_Proposal_v8_with_AI.pdf) · [editable Word](/data/private/user7/projects/Tempfile/UCMP-MB/week3_assignment/report/Week3_Streetscape_Research_Proposal_v8_with_AI.docx).

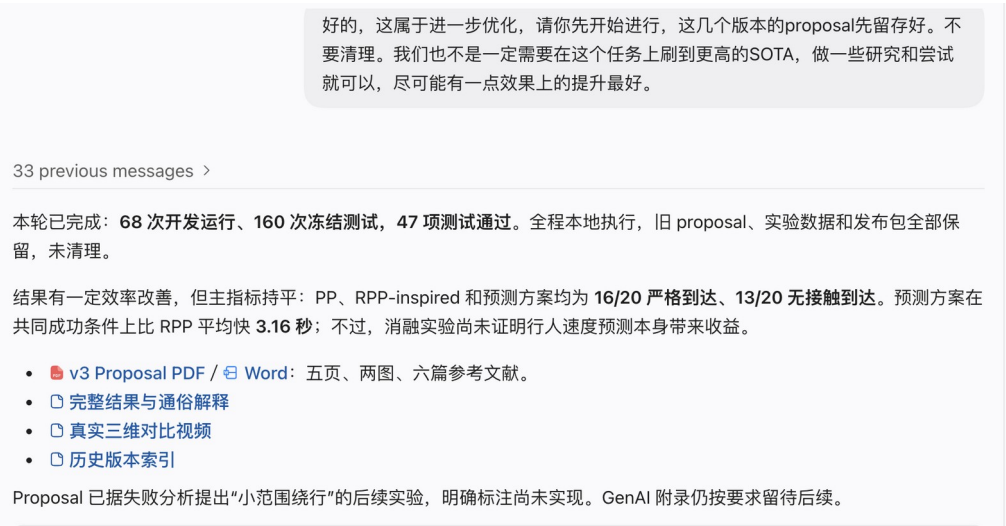
Additional Exchanges

User: “还有有些术语完全没有解释，比如说到 PPO，RPP,DWA 之类的，完全没有全称甚至。” “大量缺少逻辑关联，过渡内容，解释性内容。”

Codex: “首次出现时说明 PP、RPP、DWA、ORCA 的全称及作用，解释零速度消融为什么有必要。” “保留预测没有改善总体无接触到达率的结果；明确局部绕行尚未实现，属于下一步研究计划。”

User: “我觉得从这个 proposal 中我现在能看到问题了，但是我不知道哪些是我们做了的事情，而且后续的实验哪个是和我们相关的，我们提出来的是什么呢？哪些是拿来对比的？”

Codex: “我们已经实现的是一个沿既定路线进行短时风险预测、再选择前进速度或停车的模块；PP、RPP-inspired 和 DWA-style 是参照方法，距离让行和零速度预测用于检验模块中的不同设计。”

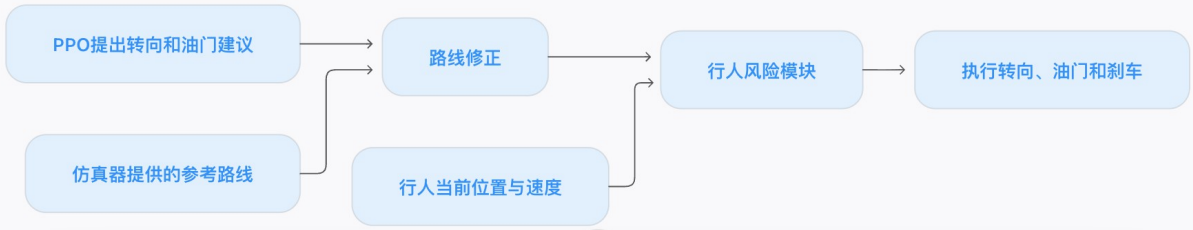


我猜肯定有一些其他的论文或者技术也解决了这个问题，请你来查找一下顺便对比一下，我觉得直接拿ppo来进行对比可能也不是一个很好的baseline，另外你说的改进部分的路线修正和新风险模块，我觉得你应该给我简单说明。我之前没有做过这一类的研究，你可以给我稍微多讲解一下

Worked for 5m 41s >

你的担心有依据：**PPO 可以保留为“现成策略迁移到这个任务”的辅助对照，但不适合作为唯一的主要 baseline。** 我查到的成熟方法已经覆盖了我们不少设计，尤其是路线跟踪、转弯限速和短期碰撞检查。当前工作的价值应定位为一个可检验的模块改进与实验，而不能仅凭胜过这份 PPO 权重就声称算法先进。

先把我们现在的系统理解成“跟路线开车”和“遇到行人时调整速度”两个问题：



系统每隔 **0.1 秒**重新观察和决策一次。

“路线修正”主要解决机器人能不能沿着给定路线走到目标。

可以想象：导航已经告诉司机前方应该左转，但司机仍想直走，或者离终点还很远就停住了。这个模块会：

- 在参考路线上寻找前方的一个跟随点；
- 判断机器人朝向、位置和转向建议是否偏离路线；
- 必要时修正转向和油门，并在转弯时限速。

这里使用了经典 **Pure Pursuit，通常译为纯追踪**的思路：朝路线前方的跟随点转向，连续重复，就能沿路线行驶。它有成熟的公开技术基础，并非我们首次提出。Coulter，1992

因此，之前说的“路线修正”实际上包含了**路线信息、转向修正、制动接管和限速这一组改动**。当前实验能说明这一组改动有效，还没有把它们各自的贡献完全拆开。

“新风险模块”主要解决沿着路线应该走多快、什么时候等待。

旧规则大致是：“预测距离太近，就减速或停车。”问题是，这样可能忽略两个事实：

- 机器人马上就会转弯，实际上不会经过行人的位置；
- 机器人停住以后，行人仍然可能走向它所在的位置。

新模块会在脑中短暂“试走”几种速度，例如停车、0.5、1、2、3 m/s，看看未来约 **1 秒**会发生什么。它考虑前方路线、加减速，以及机器人和行人的矩形占地范围，再选出预测间距足够的速度。

举个例子：行人正在从右向左穿过。机器人可能需要提前等待，也可能应该保持速度先通过；“一律停车”未必是结果更好的